

TARUNI MALLU

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Education

BITS Pilani Hyderabad Campus

B.E. Civil Engineering — Minor in Computing and Intelligence – 8.3/10.0

Oct. 2022 – May 2026

Hyderabad, Telangana

Experience

Nielsen

Member of Technical Staff I

Jul. 2026 – Present

Bengaluru, Karnataka

- Architected Python backend services, REST APIs, ETL workflows, and distributed data processing pipelines for the CMI cross-media analytics platform, processing **25+ audience measurement KPIs** across TV, digital and streaming datasets for **10 markets**, with emphasis on scalable architecture, data integrity, and production reliability.
- Engineered asynchronous data pipelines handling **50 concurrent requests per market** using configurable worker pools, retry policies, circuit breakers, job orchestration, and CI/CD workflows in Linux-based production environments, improving throughput, fault tolerance, scalability, and reliability.
- Owned end-to-end feature development across Python backend services and React/Redux interfaces, debugging API, data-flow, and distributed processing issues and delivering production features across application layers.

Nielsen

Software Engineer Intern

Jul. 2025 – Dec. 2025

Hyderabad, Telangana

- Built Python-based ETL pipelines across **10+ datasets**, designing modular transformations, automated validation, fault-tolerant processing, SQL-based data quality checks, and scalable ingestion workflows; debugged pipeline failures and improved reliability, maintainability, and performance of production data systems.
- Developed and validated the ClearDecisions Open API, implementing API testing, data validation, error handling, and SQL auditing across **1M+ staging and production records** to strengthen data integrity and service reliability.

ProvenTech Consulting Pvt. Ltd.

Software Engineer Intern

May 2024 – Jul. 2024

Hyderabad, Telangana

- Developed a production-oriented data processing and prediction platform, Dockerizing Streamlit services and implementing automated deployment workflows and scalable execution, reducing deployment time by **75%** and improving production reliability.

Projects

Port Scheduler : Multi-Process Resource Allocation Simulation | C, Linux, System V IPC, GDB, Valgrind May 2025

- Developed a discrete-event resource scheduling system in C on Linux, coordinating independent scheduler, generator, and solver processes using **System V shared memory and message queues** for inter-process communication.
- Implemented priority-based resource allocation, distributed authentication search, signal handling, and dynamic memory management; used **GDB and Valgrind** to debug process coordination, memory errors, and IPC failures.

Production Monitoring & Diagnostics System | Prometheus, Grafana, Python

October 2025

- Developed a production monitoring platform for real-time services, tracking latency, error rates, data quality, resource behavior, and system health to identify performance and reliability issues.
- Implemented automated monitoring, alerting, validation, and health evaluation workflows using Prometheus and Grafana, analyzing performance trends to detect anomalies and reduce system degradation risk by **30%**.

Data Processing & Inference Platform | Python, FastAPI, REST APIs

September 2025

- Developed a production-oriented data processing platform using Python, implementing modular pipelines, input validation, fault handling, performance evaluation, and reliable processing across large-scale workloads.
- Built a FastAPI service with REST APIs, concurrent request handling, CI/CD workflows, logging, and production monitoring, enabling low-latency processing under concurrent workloads.

Technical Skills

Systems & Fundamentals: Operating Systems, Data Structures, Algorithms, Process Management, Inter-Process Communication, Concurrency, Linux, Signal Handling, Debugging, Performance Analysis, Performance Optimization

Languages: C, C++, Python, Java, SQL

Software Engineering: Object-Oriented Programming, Software Design, Automated Testing, Error Handling, Code Quality, Fault-Tolerant Systems, Production Systems

Backend & Distributed Systems: REST APIs, FastAPI, Microservices, Distributed Systems, Asynchronous Processing, Concurrent Processing, Scalable Architecture, API Design

Data & Diagnostics: Data Processing, Data Validation, Data Quality, SQL Optimization, Data Analysis, Monitoring

Cloud & DevOps: AWS, Docker, Kubernetes, Git, GitHub, CI/CD, Jenkins, Containerization, Monitoring & Observability, Prometheus, Grafana